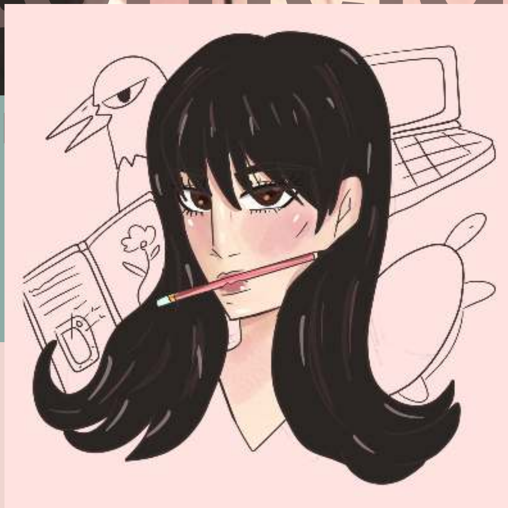

(PROBABLY ALMOST)

EVERYTHING ON THE SYLLABUS

XBI11 IAL EDEXCEL BIOLOGY

UNIT 1 & 2

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Biomolecules

1.1 Water 🚰

▼ Describe the Structure of Water:

- Contains 2 Hydrogen Atoms + 1 Oxygen Atom
- Dipole - uneven distribution of charges
- Readily forms hydrogen bonds when the slightly positively charged side of water attracts negative ions vv.

▼ Outline the importance of water as a biological molecule

- *Cohesion: The attraction between molecules of the same type*

Water is Cohesive: Since water is dipole (uneven distribution of charges) it readily forms hydrogen bonds with other water molecules. This causes them to become cohesive (clump/stick together), allowing them to flow and transport substances.

Useful in making blood flow and transpiration pull.

Water is a solvent: Water's dipole nature allows it to attract differently charged ionic compounds. The positively charged end of water attracts negative ions and the negatively charged end of water attracts the positive ions, causing them to be surrounded by water (hydrated) and dissolved.

Useful in transporting ions dissolved in the water in blood plasma.

- *Hydrophillic: Polar substances that dissociate in water/repel water*

Water has a high thermal capacity: Water is formed by many hydrogen bonds between many water molectules which means it takes a lot of energy to break up the covalent bonds - giving them a high heat capacity.

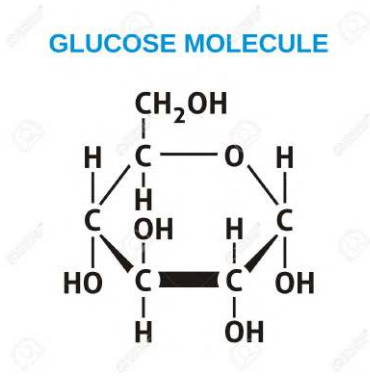
Useful in regulating temperatures to maintain homeostasis.

1.2 - 1.4 Carbohydrates 🍞

▼ Describe the structure of Monosaccharides:

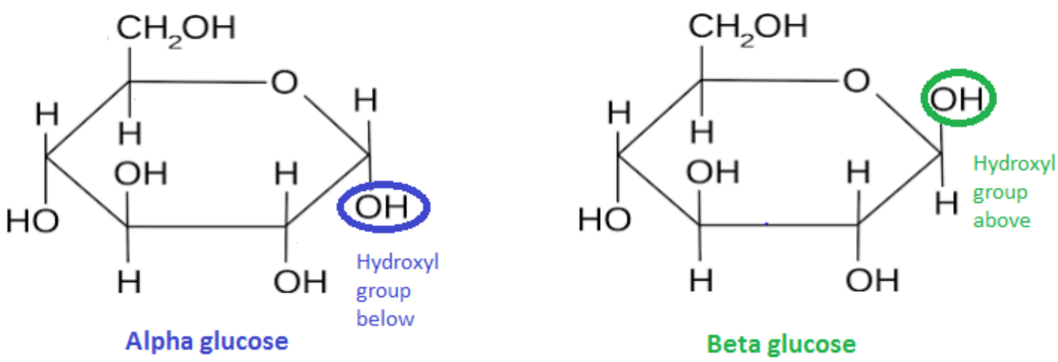
- Monomer building blocks of large carbohydrates
- Contains C,H,O
- General formula - (CH2O)N.
- Ex. Glucose, Fructose, and Galactose

▼ Describe the structure and function of Glucose



- Hexose sugar with 6 carbon, 12 hydrogen and 6 Oxygen atoms.
- Small and soluble → allows it to be dissolved into the blood and transported to cells
- Monomer building blocks for starch and glycogen
- Primary source of energy in the form of ATP in respiration
- Two types: Alpha & Beta Glucose

Alpha Glucose V. Beta Glucose



Differences:

1. Alpha glucose's hydroxl group on carbon 1 is attached below the ring, Beta Glucose's is above the ring.
2. Alpha glucose polysaccharides form helix structures, Beta glucose forms a straight chained structure.
3. Alpha glucose used for the store of energy, Beta glucose used for structural purposes.

▼ Explain how disaccharides are formed

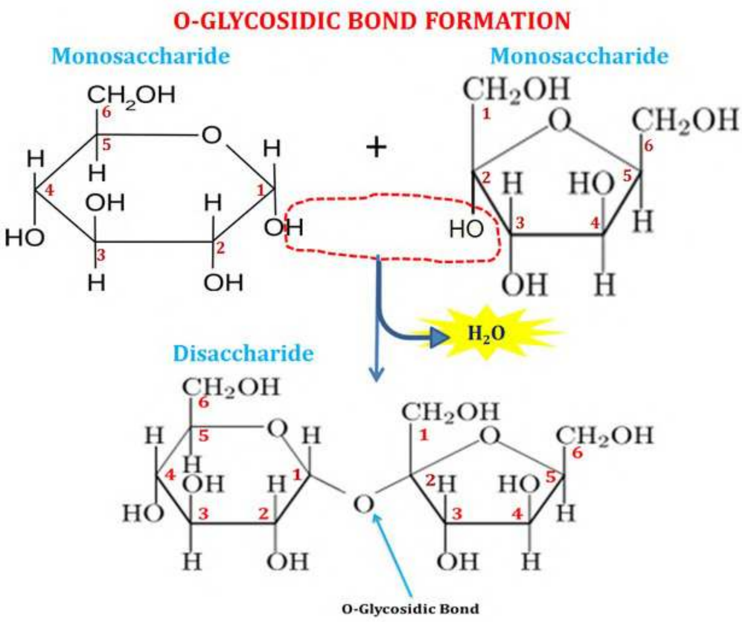
Disaccharides:

- Formed when two sugar units are joined together by a glycosidic bond between the oxygen of one monosaccharide to the carbon of another, in a condensation reaction with the removal of water.
- Soluble carbohydrates.

Maltose = alpha glucose + alpha glucose, has 1,4 glycosidic bond

Sucrose = alpha glucose + Fructose

Lactose = glucose + galactose



▼ Explain Hydrolysis

Process by which disaccharides (and polysaccharides) are broken down into monosaccharides by the addition of water which breaks their glycosidic bonds.

▼ Describe the structure and functions of polysaccharides

Polysaccharides:

- Large carbohydrate molecules formed in a series of condensation reactions linking the oxygen atom of one monosaccharide to the carbon atom of the next with a glycosidic bond, with the removal of water.
- Large and insoluble

Starch

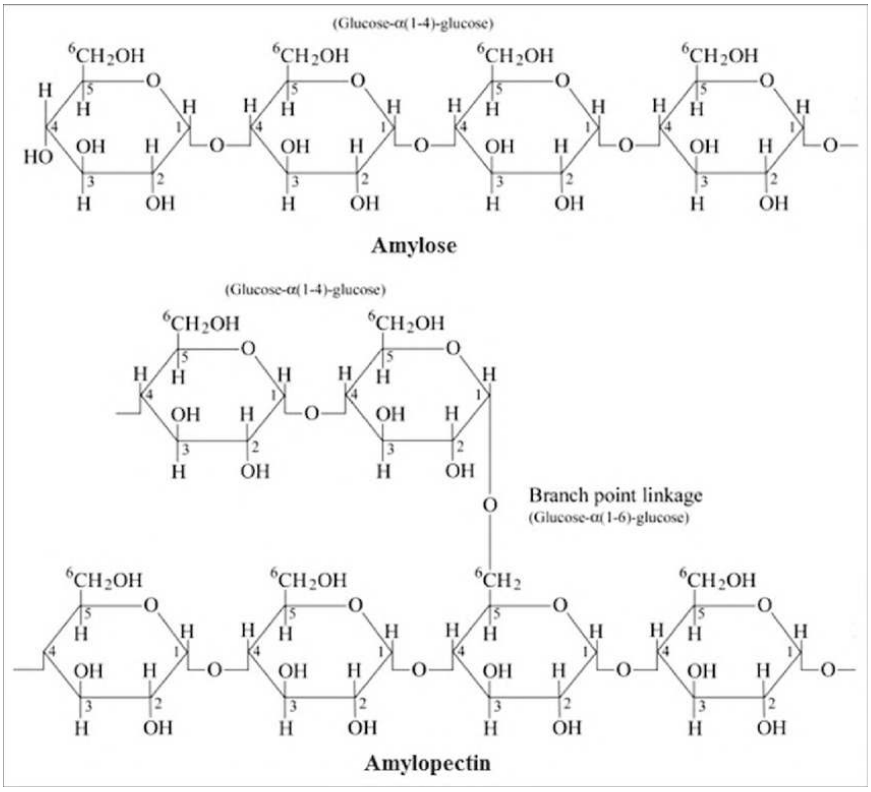
Starch = amylose + amylopectin

Amylose -

1. **Alpha glucose** molecules linked by 1,4 glycosidic bonds → Energy
2. **Compact Helix Structure** → more energy stored in a small amount of space → good energy store
3. **Unbranched** → no accessible ends where their glycosidic bonds can be broken down → good energy store

Amylopectin -

1. Alpha glucose molecules linked by 1,4 and 1,6 glycosdic bonds → Energy
2. Branched → Accessible ends where it can be easily broken down in hydrolysis → provides energy



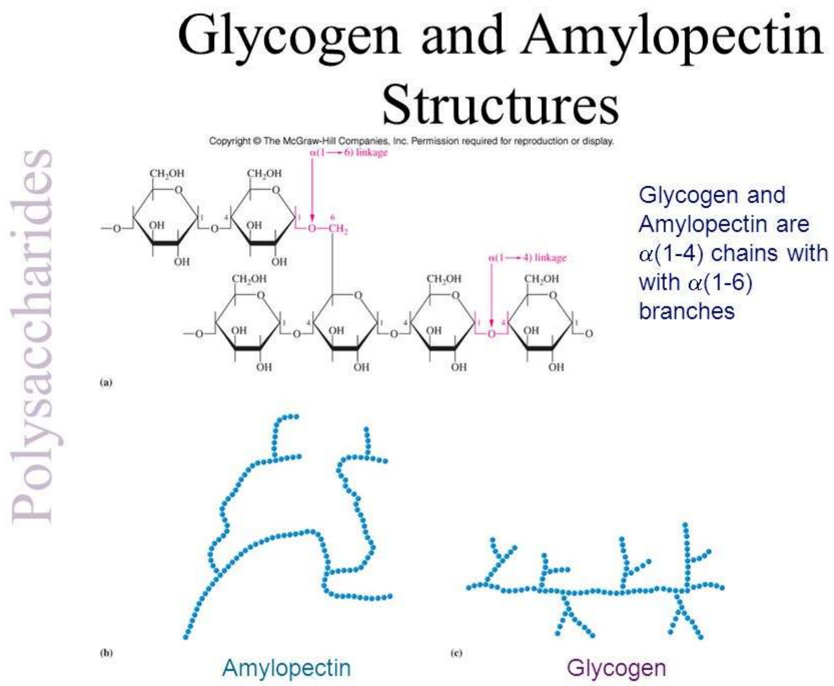
▼ Why is starch a good store of energy and metabolic substrate?

1. Compact → More energy stored in a small space
2. Insoluble → Osmotically inactive, so doesn't impact the water potential of organisms.
3. Large → too large to be lost through membranes/inactive
4. Easily hydrolysed → contain many branches with accessible ends where glycosidic bonds can be broken down to form alpha glucose molecules for respiration. **Makes it a good respiratory/metabolic substrate.**
5. Glucose polymer → glucose can be broken down to supply energy

▼ Describe the formation, structure and function of glycogen:

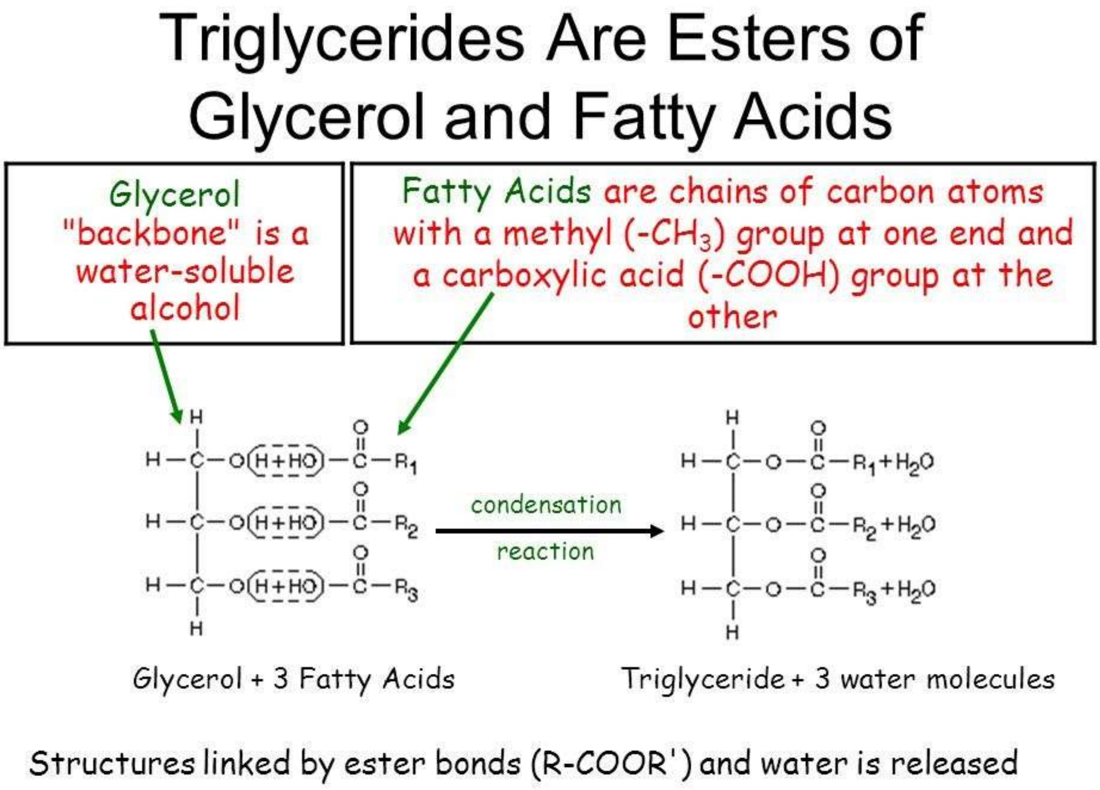
Glycogen - **Energy store in animals as animals have a high metabolic requirement and glycogen can be hydrolysed fast into glucose to provide energy for respiration.**

- 1,4 & 1,6 glycosidic bonds between alpha glucose molecules
- Many branches (more than than amylopectin)



1.5 Lipids 🛢️

- Describe the structure of a lipid
  - Large organic macromolecule containing carbon, hydrogen and oxygen.
  - Non-polar, so it doesn't dissolve in water.
  - Examples: Phospholipids and Triglycerides
- Describe the formation of triglycerides:
  - One glycerol molecule (alcohol) + 3 fatty acids in a series of condensation reactions.
  - Forms 3 ester bonds between the hydroxyl groups of the glycerol and fatty acids.



▼ What are the differences between saturated and unsaturated lipids?

1. Unsaturated lipids contain Carbon-Carbon double bonds whereas saturated contains only single bonds.
2. Unsaturated liquid at room temp., saturated is solid at room temp.
3. Unsaturated has a bent hydrocarbon chain, saturated has a straight chain.

Unsaturated lipids contain double bonds between the carbon atoms in the hydrocarbon chain. They are liquid at room temperature because their double bonds cause the fatty acid molecules to bend, which pushes the triglyceride molecules apart, weakening their intermolecular forces.

This PDF is a Compilation of Notes I (@attorneighhh) Created in my AS Year. It contains information from a wide range of sites and textbooks, including, but not limited to:

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